A SHORT CLASSIFICATION OF COMPLEX RANK-TWO GROTHENDIECK RINGS

Abstract. We give a short proof that a complex rank-two fusion category has Grothendieck ring $X^2 = 1 + nX$ with $n = 0$ or 1. A norm argument in its sphericalization first establishes sphericality without assuming pivotality. The center's Gauss sum then excludes $n \geq 2$ by a two-term triangle inequality. The argument uses standard structural results for fusion categories and does not classify tensor structures.

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1. Introduction

Ostrik classified the complex fusion categories of rank two, including their tensor structures [4]. We give a compact proof at the level of Grothendieck rings. The new proof separates the existence of a spherical structure from the fusion-rule exclusion and does not assume a braiding on the original category. For a fusion category with simples $\mathbf{1}, X$, rigidity gives $X^* = X$ and

$$X^2 = \mathbf{1} + nX, \qquad n \in \mathbb{Z}_{\geq 0}. \tag{1}$$

Write K_n for this based ring.

Theorem 1.1. *For a fusion category $\mathcal{C}$ over $\mathbb{C}$ with exactly two simple objects, $K_0(\mathcal{C})$ is isomorphic to K_0 or K_1. Every such category admits a spherical structure. Both rings occur, respectively in a pointed category and in a Fibonacci category.*

The statement concerns fusion rings, rather than the monoidal-equivalence classification of their realizations. In particular, the two categories with ring K_0 and the two with ring K_1 in [4] need not be identified to prove Theorem 1.1.

2. A spherical structure from a norm

We recall the general results used below. A fusion category over $\mathbb{C}$ admits a *sphericalization* $\pi : \mathcal{S} \to \mathcal{C}$, where $\mathcal{S}$ is spherical. Every simple V has two simple lifts V_+, V_-; the lifts of $\mathbf{1}$ are $\mathbf{1}$ and a central invertible object ε of dimension -1, and $\varepsilon \otimes V_+ = V_-$ [1, Remark 3.1]. If V is self-dual, so are both lifts [5, Section 5.1]. We use the general *formal-codegree divisibility theorem*: if f is a formal codegree of a complex fusion category $\mathcal{A}$, then $\dim(\mathcal{A})/f$ is an algebraic integer [5, Corollary 2.14]. The global dimension here is the intrinsic global dimension and does not require a pivotal structure.

Lemma 2.1. *Suppose $n \geq 1$ in (1). If x is either lift of X in the sphericalization, its signed fusion rule modulo $\varepsilon = -1$ is*

$$x^2 = 1 + \widetilde{n}x, \qquad \widetilde{n} \in \mathbb{Z}, \quad |\widetilde{n}| \leq n, \quad \widetilde{n} \equiv n \pmod 2. \tag{2}$$

Moreover $|\widetilde{n}| = n$.

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Proof. As x is self-dual, its square contains $\mathbf{1}$ exactly once and ε zero times. The other constituents have the form $rx + s\varepsilon x$ with $r, s \geq 0$ and $r + s = n$. Reducing modulo $\varepsilon = -1$ gives $\widetilde{n} = r - s$, proving the first assertions.

Let $d = \dim(x)$. The spherical dimension character kills $\varepsilon + 1$, so $d^2 = 1 + \widetilde{n}d$. The global dimension of $\mathcal{C}$ equals that of one choice of lift squared, namely

$$\delta = 1 + d^2; \tag{3}$$

the global dimension of $\mathcal{S}$ is 2δ. Put $R = \mathbb{Q}[t]/(t^2 - nt - 1)$. Its two formal codegrees are $f_\pm = 1 + u_\pm^2$, where $u_\pm = (n \pm \sqrt{n^2+4})/2$. Consequently

$$N_{\mathbb{Q}(\sqrt{n^2+4})/\mathbb{Q}}(f_+) = n^2 + 4. \tag{4}$$

The number $n^2 + 4$ is nonsquare for $n \geq 1$, so R is a quadratic field. The two lifts of each simple double each codegree of R in $K_0(\mathcal{S})$. Thus $2f_+$ is a formal codegree of $\mathcal{S}$, and general codegree divisibility says that δ/f_+ is an algebraic integer.

To take its norm in a single quadratic field, we recall a second general formal-codegree fact: the codegrees of a spherical category belong to the field generated by its simple dimensions [5, Corollary 2.15]. This field is $L = \mathbb{Q}(d)$, of degree at most two, and it contains both roots $2f_\pm$ (the conjugate codegrees of $\mathcal{S}$). Hence $L = \mathbb{Q}(\sqrt{n^2+4})$; it is also the field of $\widetilde{n}$'s quadratic character, so $\widetilde{n} \neq 0$. The conjugate of δ is $1 + d'^2$ with d' the other root of $t^2 - \widetilde{n}t - 1$. Taking norms in L yields

$$N_{L/\mathbb{Q}}(\delta/f_+) = \frac{\widetilde{n}^2 + 4}{n^2 + 4} \in \mathbb{Z}_{>0}.$$

The ratio is positive and at most one by $|\widetilde{n}| \leq n$. Therefore it equals one and $|\widetilde{n}| = n$. □

Corollary 2.2. *Every complex rank-two fusion category admits a spherical structure.*

Proof. For $n = 0$ the category is pointed and hence pseudo-unitary; its canonical pivotal structure is spherical [1, Propositions 8.23–8.24]. For $n \geq 1$, changing x to εx changes $\widetilde{n}$ to $-\widetilde{n}$. By Lemma 2.1 we may choose $\widetilde{n} = n$. Since $r + s = n$ and $r - s = n$, we have $s = 0$: $x^2 = \mathbf{1} \oplus nx$ already inside $\mathcal{S}$. The full replete subcategory with simples $\mathbf{1}, x$ is closed under tensor products and duals. The forgetful tensor functor π restricts to a tensor equivalence from it to $\mathcal{C}$. Restricting and transporting the spherical structure of $\mathcal{S}$ proves the claim. □

3. The center's two known summands

Let $\mathcal{C}$ now have the spherical structure of Corollary 2.2, write $d = \dim(X)$, and set $D = \dim(\mathcal{C}) = 1 + d^2$. We recall two standard properties of its modular Drinfeld center $\mathcal{Z}(\mathcal{C})$:

$$\sum_{B \in \operatorname{Irr} \mathcal{Z}(\mathcal{C})} \dim(B)^2 = D^2, \qquad \sum_{B \in \operatorname{Irr} \mathcal{Z}(\mathcal{C})} \theta_B \dim(B)^2 = D. \tag{5}$$

The first follows from $\dim \mathcal{Z}(\mathcal{C}) = D^2$ and the second is the center Gauss-sum identity [3, Theorem 1.2]. For completeness, the second identity follows by writing $\dim(B) = \sum_V [F(B) : V] \dim(V)$, using adjunction, and summing $\dim(V) \operatorname{Tr}(\theta_{I(V)})$ over the simple objects V: the twist trace is D for $V = \mathbf{1}$ and zero otherwise [5, Theorem 2.5]. Here θ_B is the canonical twist; its values are roots of unity by the Vafa theorem.

Let $F : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ be forgetful and I its right adjoint. The standard formula $FI(\mathbf{1}) = \bigoplus_V V \otimes V^*$ gives $FI(\mathbf{1}) = 2\mathbf{1} \oplus nX$. By adjunction, $\dim_{\mathbb{C}} \operatorname{End}_{\mathcal{Z}(\mathcal{C})} I(\mathbf{1}) = 2$. The simple unit occurs once in $I(\mathbf{1})$, so semisimplicity implies

$$I(\mathbf{1}) = \mathbf{1} \oplus A, \qquad F(A) = \mathbf{1} \oplus nX, \qquad \dim(A) = 1 + nd = d^2. \tag{6}$$

The twist of each summand of $I(\mathbf{1})$ is one by the same trivial-twist theorem [5, Theorem 2.5]; equivalently, the canonical connected commutative separable algebra $I(\mathbf{1})$ has trivial twist.

Proof of Theorem 1.1. The sphericality assertion is Corollary 2.2. Assume $n \geq 1$. The two roots of $d^2 = 1 + nd$ are $d_\pm = (n \pm \sqrt{n^2+4})/2$. Choose an automorphism σ of $\overline{\mathbb{Q}}$ sending the actual categorical dimension d to $d_+ > 0$. For any simple central object B, its dimension is the dimension of $F(B) = a_B\mathbf{1} \oplus b_B X$ with $a_B, b_B \in \mathbb{Z}_{\geq 0}$, and hence $\sigma(\dim(B)) = a_B + b_B d_+ > 0$. Applying σ to (5) preserves the equalities and sends each twist to another root of unity; no Galois descent of associators is involved.

Set $t = d_+^2$ and $D_+ = 1 + t$. The two objects in (6) contribute $1 + t^2$ to both identities in (5). Thus the other simple central objects satisfy

$$\sum_{B \neq \mathbf{1}, A} \sigma(\dim(B))^2 = D_+^2 - 1 - t^2 = 2t, \qquad \sum_{B \neq \mathbf{1}, A} \sigma(\theta_B)\sigma(\dim(B))^2 = D_+ - 1 - t^2 = t(1-t).$$

All coefficients in the first sum are nonnegative real numbers and $|\sigma(\theta_B)| = 1$. The triangle inequality gives $t|1 - t| \leq 2t$, so $t \leq 3$. For $n \geq 2$, $d_+ > n \geq 2$ and therefore $t > 4$, a contradiction. Hence $n \leq 1$. Finally K_0 is realized by $\mathrm{Vec}_{\mathbb{Z}/2}$ and K_1 by the Fibonacci fusion category, for instance the even part of the $SU(2)_3$ fusion category [4]. Both are spherical. □

Remark 3.1 (Positive characteristic). The argument above is specific to characteristic zero. Etingof, Ostrik and Venkatesh [2, Propositions 4.6–4.7] prove that over an algebraically closed field of positive characteristic no *braided* fusion category realizes K_n for $n \geq 2$, and explicitly leave the general fusion-category case open. In particular their result cannot be used to assert a full positive-characteristic classification. A category of zero global dimension in characteristic p need not have a semisimple modular center, so the proof of (5) does not carry over directly. We know of no later general resolution of the nonbraided rank-two problem in positive characteristic.

AI disclosure. This manuscript, including its arguments and supporting calculations, was generated with AI assistance. Independent human verification and Lean formalization are pending.

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